Incidence and timing of postoperative complications after total hip and knee arthroplasty

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Background: Follow-up protocols after total hip or knee arthroplasty (THA or TKA, respectively) have little uniformity, which can lead to emergency department (ED) visits for postoperative complications. We sought to determine the incidence and timing of postoperative complications after THA or TKA.

Methods: We conducted a population-based retrospective cohort study of all adults in Ontario who underwent primary THA or TKA between 2010 and 2019. We used data available through ICES. We identified medical and surgical complications, ED visits, and hospital readmissions using institutional databases and Ontario Health Insurance Plan claims. Outcomes included major medical complications within 30 days and surgical complications within 1 year after surgery.

Results: We included 158 503 and 103 728 patients who underwent TKA and THA, respectively. The incidence of medical complications within 30 days was 2.90% after TKA and 2.42% after THA. Visits to the ED (20.0% after TKA, 16.9% after THA) and readmission rates within 30 days (3.8% after TKA, 4.1% after THA) were similar for both groups. Visits to the ED occurred at a median of 10 days after surgery for both groups, with readmissions at a median of 12 and 13 days after TKA and THA, respectively. The incidence of major TKA complications was 1.6%, with a median time of 84 (interquartile range [IQR] 26–224) days. The incidence of major THA complications was 2.2%, with a median time of 29 (IQR 16–80) days.

Conclusion: Our findings suggest follow-up contact 7–10 days after THA or TKA to minimize ED visits, with at least 1 subsequent in-person follow-up at 5–6 weeks after surgery. After that, surgeons may personalize additional follow-ups as needed.

Contexte : Les protocoles de suivi après une arthroplastie totale de la hanche (ATH) ou du genou (ATG) manquent d'uniformité, ce qui peut entraîner des complications postopératoires nécessitant une visite aux services des urgences (SU). Nous avons cherché à déterminer l'incidence et le moment des complications à la suite d'une ATH ou d'une ATG.

Méthodes : Nous avons mené une étude de cohorte rétrospective basée sur la population de l'ensemble des adultes en Ontario ayant subi une ATH ou une ATG primaire entre 2010 et 2019, identifiés à partir des données de l'ICES. En parcourant les bases de données d'établissements et des demandes de remboursement du Régime d'assurance-santé de l'Ontario, nous avons répertorié les complications médicales et chirurgicales, les visites aux SU et les réhospitalisations. Les paramètres comprenaient les complications médicales majeures dans les 30 jours et les complications chirurgicales dans l'année suivant l'opération.

Résultats : Nous avons inclus 158 503 personnes ayant subi une ATG, et 103 728 ayant subi une ATH. L'incidence des complications médicales dans les 30 jours était de 2,90 % après l'ATG et de 2,42 % après l'ATH. Les taux de visites aux SU (20,0 % après l'ATG, 16,9 % après l'ATH) et de réhospitalisation dans les 30 jours (3,8 % après l'ATG, 4,1 % après l'ATH) étaient semblables pour les 2 groupes. La médiane du moment de la visite aux SU était de 10 jours après l'opération pour les 2 groupes, tandis que la médiane du moment de la réhospitalisation était de 12 jours après l'ATG et 13 jours après l'ATH. L'incidence des complications majeures de l'ATG s'élevait à 1,6 %, avec un intervalle médian de 84 (écart interquartile [ÉI] 26–224) jours. L'incidence des complications majeures de l'ATH s'élevait à 2,2 %, avec un intervalle médian de 29 (ÉI 16–80) jours.

Conclusion : Nos résultats suggèrent qu'un suivi 7–10 jours après une ATH ou une ATG permettrait de réduire les visites aux SU, avec au moins 1 consultation en personne 5–6 semaines après l'opération. Les chirurgiens et chirurgiennes pourront ensuite adapter le suivi aux besoins.

Recent measures regarding total joint arthroplasty have focused on decreasing health care costs without sacrificing the quality of care and patient outcomes.[1–3] Total hip and knee arthroplasty (THA and TKA, respectively) are among the most common surgical procedures. Although evidence-based approaches to performing these surgeries exist, there is less uniformity in follow-up protocols after surgery, particularly in the first year. Some surgeons follow a structured protocol, whereas others instruct patients to call as needed. This approach can have the unfortunate effect of delaying presentation for complications or unnecessary presentation to the emergency department (ED).

Healthcare Excellence Canada developed a clinical pathway for enhanced recovery after inpatient and outpatient THA and TKA.[4] This clinical pathway aims to provide practitioners in Canada with evidence-based strategies to improve surgical outcomes among patients who undergo total joint arthroplasty. This strategy was developed by a multidisciplinary national group of clinicians experienced in hip and knee surgery. The Enhanced Recovery After Surgery (ERAS) consensus guideline for perioperative care suggests a follow-up strategy for all surgeons.[5] For outpatient cases, a nurse may visit the patient at home in the first few days after surgery to check the wound, dressing, and vital signs. Afterward, follow-up visits should occur as needed in the first weeks after surgery, according to the standard of care in the area.

Decreasing readmission rates after THA and TKA would reduce overall costs, health care burden, and postoperative morbidity.[1–3] Hospital readmission rates after THA and TKA are a severe financial problem.[2,6,7] Consequently, the Centers for Medicare and Medicaid Services developed the Hospital Readmissions Reduction Program in the past decade. This program aims to fine hospitals for high readmission rates for certain conditions; THA and TKA have been included since 2014. Nevertheless, the expansion of the Hospital Readmissions Reduction Program has not led to a significant decrease in readmission rates.[6–9]

Hospital readmission rates and ED visits after THA or TKA have previously been studied in privatized health care systems, but fewer studies have investigated rates in publicly funded, single-payer systems. In 2020, Ravi and colleagues[10] conducted a population-based retrospective cohort study using health administrative databases to analyze the prevalence of ED visits after elective THA or TKA in Ontario between April 2011 and March 2016. They identified 42 273 patients who underwent THA and 70 725 who underwent TKA, of whom 5640 (13.3%) and 11 224 (15.9%), respectively, presented to the ED within 30 days after surgery. Fewer than 1% required readmission, and most visits were for wound issues, infection, and pain concerns.[10] Similarly, Ross and colleagues[11] reviewed data from all 205 152 TKAs performed in Ontario between 2003 and 2016 to determine readmissions and ED visits. All-cause 30-day readmissions did not change significantly over the study period, but ED visits increased significantly. In total, 31 198 (15.2%) patients visited the ED within 30 days of TKA, with only 20.1% of these readmitted. Again, the most common reasons for ED visits within 30 days were surgical site infections and wound complications (20.8%), pain and swelling (13.6%), bleeding or anemia (5.4%), and venous thromboembolism (4.7%).[11]

Although several studies have been published on identifying predictors of medical and surgical complications after surgery, few have looked at the typical timing of these complications to inform follow-up practices. We sought to identify the incidence and timing of several common complications after THA and TKA to suggest a probable follow-up strategy to avoid unnecessary ED visits.

Methods

Study sample

We conducted a descriptive, population-based retrospective cohort study. We defined a cohort of consecutive adults (aged > 18 yr) who received their first primary elective THA and TKA for osteoarthritis between Jan. 1, 2010, and Dec. 31, 2019. Ontario has a publicly funded health care system, with no private coverage for primary or revision THA and TKA, allowing for complete identification of all hip and knee replacements and any subsequent complications managed in this province.

Data sources

We used hospital discharge abstracts from the Canadian Institute for Health Information (CIHI) Discharge Abstract Database and physician claims from the Ontario Health Insurance Plan (OHIP). We identified patients using specific procedure and diagnostic codes (CIHI codes 1VA53LA and 1VH53LA; OHIP codes R440 and R441) from the Canadian version of the *10th revision of the International Statistical Classification of Diseases* (ICD-10-CA) and the *Canadian Classification of Health Interventions* (CCI). Patient demographics, including age and sex, were accessed via the OHIP Registered Persons Database (RPDB),[12–14] using the same validated definitions as the Ministry of Health and Ministry of Long-Term Care.[15]

Outcomes

We examined the occurrence of major medical complications, such as acute myocardial infarction (AMI), deep vein thrombosis (DVT), pulmonary embolism (PE), pneumonia, return to the ED within 30 days of surgery, readmission to hospital within 30 days of surgery and of major surgical complications, including revision arthroplasty, periprosthetic joint infection (PJI) requiring additional surgery, and dislocation requiring open or closed reduction within 1 year of surgery. Revision procedures were identified using ICD-10-CA/CCI procedure codes accompanied by the supplementary status attribute "R," with an appropriate surgeon billing code. We identified infections requiring surgery by the first occurrence of an ICD-10-CA diagnostic code for intra-articular infection, with a confirmatory code for irrigation and débridement or an OHIP fee code for a spacer insertion. We identified dislocations by the presence of a procedure code for closed or open hip reduction.

Covariates

We measured several factors associated with the occurrence of complications after THA and TKA. These included patient age, sex, socioeconomic status, and comorbidities. We categorized comorbidities listed on hospital discharge abstracts in the 3 years before the index admission according to an adaptation of the Charlson Comorbidity Index.[16] We used Johns Hopkins Adjusted Clinical Groups, based on diagnosis codes from hospital admissions and physician visits in the 2 years before the index admission, to classify patients as frail.[17] We identified the presence of diabetes mellitus using a validated algorithm.[18] We used the neighbourhood income quintile as a surrogate measure for socioeconomic status and living conditions. This categorizes small geographic areas into 5 roughly equal population groups, with the lowest quintile referring to the least affluent neighbourhoods.[19,20]

Statistical analysis

We reported continuous variables as medians and interquartile ranges (IQRs) and categorical variables as frequencies and percentages. All analyses were performed at ICES using SAS version 9.3 for UNIX (SAS Institute). The type I error probability was set to 0.05 for all analyses.

Ethics approval

Institutional review board permission is included in the administrative data use agreement from ICES. Section 45 of the Ontario *Personal Health Information Protection Act* authorizes the use of the data and does not require independent review by an ethics board. This study was approved by the University of Toronto and Affiliated Teaching Hospitals and Health Sciences Human Research Ethics Board.

Results

We identified 262 231 procedures (158 503 TKAs and 103 728 THAs) to include in the analysis. The median age was 68 (IQR 61–74) years and 67 (IQR 59–75) years among patients who underwent TKA and THA, respectively (Table 1). For the purposes of this study, follow-up was meant to end at 1 year after the index surgery. We analyzed the 30-day and 1-year complication and readmission rates in both groups of patients.

Thirty-day complication rate and hospital readmissions

Among patients who underwent TKA or THA, 1318 (0.8%) or 1025 (1.0%), respectively, had a delirium episode with complete resolution before being discharged. An AMI was diagnosed in 734 (0.5%) patients after TKA at a median time of 0 (IQR 0–5) days. Similarly, 492 (0.5%) patients had an AMI after THA at a median time of 0 (IQR 0–2) days. A DVT was diagnosed in 887 (0.6%) patients who underwent TKA at a median time of 9 (IQR 6–17) days. Only 429 (0.4%) patients had a DVT after THA at a median time of 11 (IQR 6–19) days. Similar rates were retrieved for PE. After TKA, 819 (0.5%) patients had a PE at a median time of 5 (IQR 0–12) days. In the THA group, 232 (0.22%) patients had a PE at a median time of 10 (IQR 4–20) days.

At a median time of 0 (IQR 0–9) days, 820 (0.5%) patients had pneumonia after TKA. Lower numbers were found after THA, with 393 (0.3%) patients at a median time of 0 (IQR 0–12) days.

Table 1: Characteristics of patients who underwent total hip arthroplasty (THA) and total knee arthroplasty (TKA) over the study period

	No. (%) of patients*	
Characteristic	TKA *n* = 158 503	THA *n* = 103 728
Age, yr, median (IQR)	68 (61–74)	67 (59–75)
Sex		
Female	97 637 (61.6)	56 280 (54.3)
Male	60 866 (38.4)	47 448 (45.7)
Income quintile		
1 (lowest)	27 452 (17.3)	16 396 (15.8)
2	32 377 (20.4)	19 580 (18.9)
3	32 719 (20.6)	20 394 (19.7)
4	32 266 (20.4)	21 523 (20.7)
5 (highest)	33 312 (21.0)	25 570 (24.7)
Comorbidities		
Diabetes	11 989 (7.6)	5958 (5.7)
Frail	11 290 (7.1)	8345 (8.0)
Charlson Comorbidity Index score		
0	107 185 (67.6)	75 934 (73.2)
1	29 808 (18.8)	15 416 (14.9)
2	13 857 (8.7)	7773 (7.5)
3	4455 (2.8)	2465 (2.4)
4	1564 (1.0)	967 (0.9)
≥ 5	1634 (1.0)	1173 (1.1)
Admission characteristics		
General anesthetic	19 797 (12.5)	16 045 (15.5)
Spinal anesthetic	132 283 (83.5)	83 591 (80.6)

IQR = interquartile range.
*Unless indicated otherwise.

Visits to the ED within 30 days after the index surgery were slightly more common among patients who underwent TKA. More precisely, 31 651 (20.0%) patients after TKA were assessed in the ED at a median time of 10 (IQR 6–16) days, compared with 17 547 (16.9%) patients after THA at a median time of 10 (IQR 6–16) days. Ultimately, both groups had similar readmission rates within 30 days. After TKA, 5974 (3.8%) patients were readmitted at a median time of 12 (IQR 7–19) days. Similarly, 4302 (4.1%) patients who underwent THA were readmitted at a median time of 13 (IQR 7–20) days.

The 30-day mortality rate was 0.2% for both TKA (n = 265) and THA (n = 179) groups, with a median time of 8 (IQR 4–16) and 10 (IQR 4–17) days, respectively.

One-year complication rate and hospital readmissions

As expected, the dislocation rate within 1 year after index surgery was higher among patients who underwent THA than among those who underwent TKA. Only 92 (0.1%) patients had a TKA dislocation at a median time of 0 (IQR 0–56) days, whereas 376 (0.4%) THA dislocations were recorded at a median time of 50 (IQR 18–134) days.

We observed similar results concerning PJI during the first postoperative year. A PJI was diagnosed in 778 (0.5%) patients after TKA at a median time of 65 (IQR 27–183) days. Similarly, 602 (0.6%) patients had a PJI after THA at a median time of 36 (IQR 20–100) days.

The revision rate for any reason within 1 year was higher after THA. At a median time of 216 (IQR 111–292) days, 775 (0.5%) TKAs were revised. On the other hand, 1239 (1.2%) THAs needed to be revised earlier, at a median time of 43 (IQR 17–151) days. Ultimately, patients who underwent THA had higher readmission rates within 1 year after index surgery. At a median time of 114 (IQR 30–244) days, the 1-year readmission rate after TKA was 0.9% (n = 1382). In contrast, 1.7% (n = 1759) of patients who underwent THA needed to be readmitted at a median time of 34 (IQR 17–111) days. The 1-year mortality rate was 1.0% (n = 1644) for TKAs and 1.3% (n = 1319) for THAs at a median time of 179 (IQR 72–283) and 172 (IQR 69–272) days, respectively. Table 2 summarizes the outcomes in both TKA and THA groups.

Discussion

In this study, we analyzed the probable period in which common complications appear after THA and TKA, with the aim of suggesting a follow-up protocol to identify and decrease the likelihood of complications and readmissions to the ED.

Carey and colleagues[21] found that 3-day readmission rates were 4% and 2.95% for outpatient TKA and THA, and 5.56% and 3.21% for inpatient TKA and THA, respectively. Our study showed similar readmission rates within 30 days after index surgery. After TKA, 5974 (3.8%) patients were readmitted at a median time of 12 (IQR 7–19) days. After THA, 4302 (4.1%) patients were readmitted at a median time of 13 (IQR 7–20) days.

Another critical variable is the exact reason for ED visits and, consequently, the reason for readmission, reoperation, or revision. Having that information available could be helpful to guide follow-up appointments. In 2020, Ravi

Table 2: Incidence and time to complications

	TKA		THA	
Complication	No. (%) of patients n = 158 503	Time to event, d, median (IQR)	No. (%) of patients n = 103 728	Time to event, d, median (IQR)
Within 30 d				
AMI	734 (0.5)	0 (0–5)	492 (0.5)	0 (0–2)
PE	819 (0.5)	5 (0–12)	232 (0.2)	10 (4–20)
DVT	887 (0.6)	9 (6–17)	429 (0.4)	11 (6–19)
Pneumonia	820 (0.5)	0 (0–9)	393 (0.3)	0 (0–12)
Return to ED	31 651 (20.0)	10 (6–16)	17 547 (16.9)	10 (6–16)
Readmission	5974 (3.8)	12 (7–19)	4302 (4.1)	13 (7–20)
Death	265 (0.2)	8 (4–16)	179 (0.2)	10 (4–17)
Any complication	26 859 (16.9)	9 (5–16)	15 463 (14.9)	9 (5–15)
Within 1 yr				
Dislocation	92 (0.1)	0 (0–56)	376 (0.4)	50 (18–134)
Revision	775 (0.5)	216 (111–292)	1239 (1.2)	43 (17–151)
Infection	778 (0.5)	65 (27–183)	602 (0.6)	36 (20–100)
Readmission	1382 (0.9)	114 (30–244)	1759 (1.7)	34 (17–111)
Death	1644 (1.0)	179 (72–283)	1319 (1.3)	172 (69–272)
Any complication	2480 (1.6)	84 (26–224)	2320 (2.2)	29 (16–80)

AMI = acute myocardial infarction; DVT = deep vein thrombosis; ED = emergency department; IQR = interquartile range; PE = pulmonary embolism; THA = total hip arthroplasty; TKA = total knee arthroplasty.

and colleagues[10] reported that 1 in 7 patients in Ontario presented to an ED within 30 days of primary THA or TKA. Moreover, well over 80% of ED visits were for family practice–sensitive conditions, including issues related to surgical wounds and pain that could have been managed as an outpatient, with fewer than 1% of patients requiring readmission. In our study, although uncommon, most medical complications requiring readmission happened in the first week after surgery. In this sense, AMI, PE, DVT, and pneumonia during the immediate postoperative period must be suspected and considered among patients with ischemic heart disease and related risk factors who undergo THA and TKA. On the other hand, most surgical readmissions (dislocation, PJI, revision arthroplasty) occurred later in the follow-up period, at a minimum of 1 month after surgery.

From the perspective of clinicians, they mainly want to see patients in follow-up to check for wound issues, early infection, or warning signs for failed implants and to confirm the progression of the range of motion. On the other hand, they are less likely to be able to identify or prevent patients from having pneumonia or further medical complications. From another perspective, orthopedic surgeons want to ensure that problems are identified early to prevent more serious issues in the long term, such as osteolysis and aseptic lymphocyte-dominant vasculitis-associated lesion.

Follow-up visits are unlikely to prevent most early postoperative medical complications, such as PE, DVT, AMI, and pneumonia, and maybe not even PJI or dislocation. In this sense, preoperative optimization may be more likely to prevent early postoperative issues. However, understanding why and when patients are being readmitted may allow surgeons to schedule follow-ups to identify complications in a timely fashion.

Limitations

As an evidence-based study, our analysis introduced heterogeneity by encompassing different institutional perioperative protocols using the most extensive provincial database available. However, our study was not without limitations, most of which have been related to the use of an administrative database. Although we used a large database, its retrospective nature may have introduced selection bias and data may have been missing from electronic records. Given its retrospective nature, we could not control for many factors that can affect results, such as smoking status. Finally, outcomes included only major medical complications within 30 days and surgical complications within 1 year of surgery. This study did not record day-to-day ED visits or analyze minor visits. As we know, most institutions that provide THA or TKA receive minor consultations (e.g., wound drainage, stitch abscesses, joint swelling, hematomas) that do not generally lead to admission or surgery. A more extended follow-up period would have been helpful for a more comprehensive analysis. Despite these limitations, our findings provide important information for the orthopedic community and primary care physicians and could be used to suggest a probable follow-up strategy to avoid unnecessary ED visits after TKA or THA.

Conclusion

The first postoperative month, especially the first week, is the most crucial time for medical complications after THA and TKA. Based on our findings, THA and TKA should be carefully indicated among patients with multiple comorbidities and risk factors. For surgical complications, the highest risk was recorded around the first postoperative month, but high risk continued for the second, third, and even fourth months after surgery. Therefore, we suggest an initial virtual or in-person contact 7–10 days after surgery and a second in-person follow-up visit around the first postoperative month, with additional visits scheduled depending on the patient's requirements for further counselling and education. By proposing this strategy, we aim to give orthopedic surgeons the ability to anticipate and reduce rates of complications and readmission.

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Competing interests: Mark Spangehl reports holding stock in Sonoran Biosciences. No other competing interests were declared.

Contributors: Bheeshma Ravi conceived and designed the work. Fernando Diaz-Dilernia, Yaniv Steinfeld, Daniel Pincus, and Mark Spangehl contributed to data acquisition, analysis, and interpretation. Fernando Diaz-Dilernia, Yaniv Steinfeld, and Bheeshma Ravi drafted the manuscript. All of the authors revised it critically for important intellectual content, gave final approval of the version to be published, and agreed to be accountable for all aspects of the work.

Data sharing: The administrative data set for this study is held securely in coded form at ICES. Although legal data-sharing agreements between these data stewards and data providers prohibit ICES from making the data set publicly available, access may be granted to those who meet prespecified criteria for confidential access. Readers are welcome to contact the research team for further information. Data queries to ICES can be directed to Data and Analytic Services at das@ices.on.ca.

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